

Task 3

Protein Structure Prediction & Visualization (PyMOL)

Aim

To predict, model, and visualize the three-dimensional tertiary structure of human KRAS protein using computational tools and analyze key structural features in PyMOL.

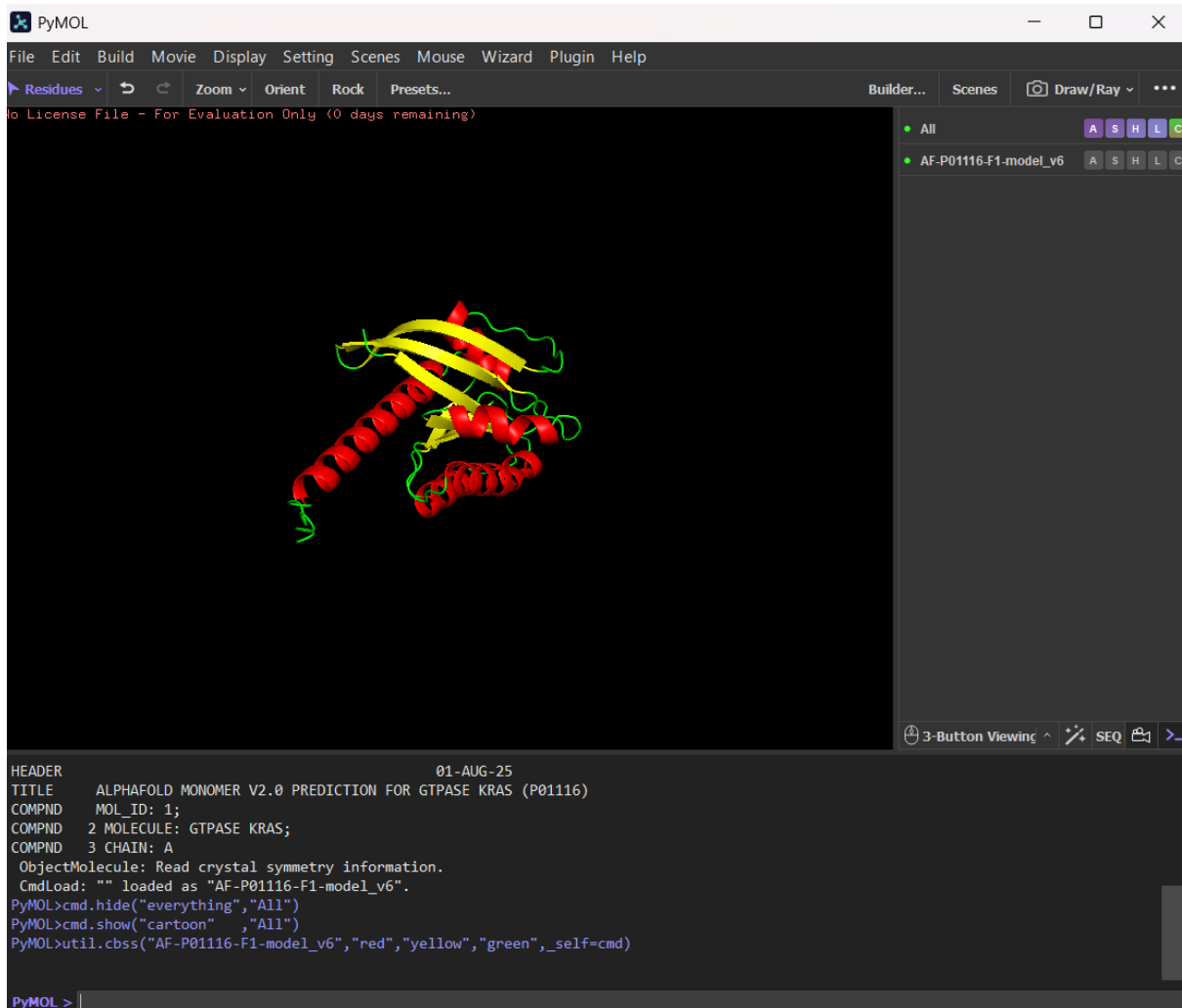
Introduction

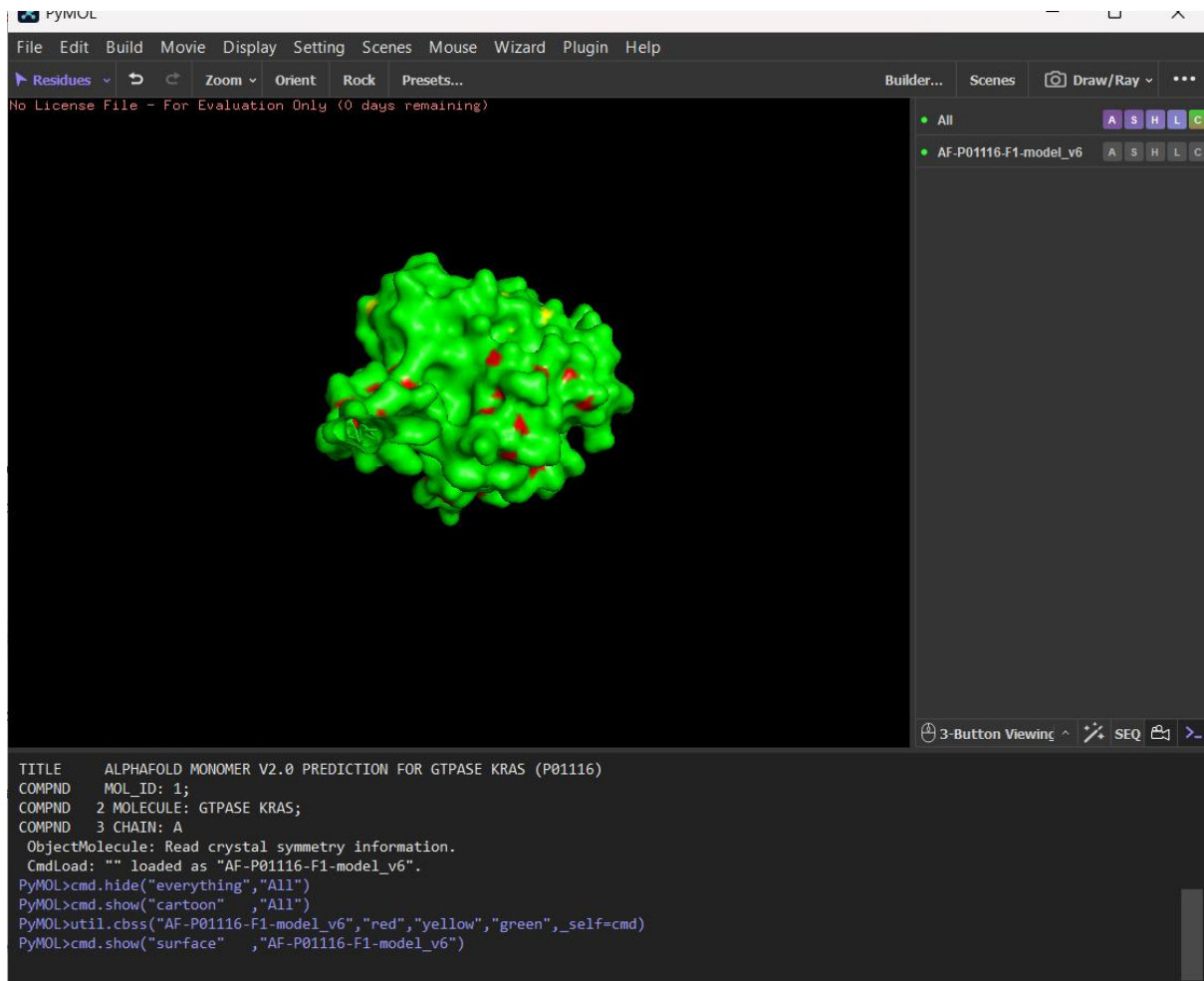
Understanding the three-dimensional structure of a protein is essential for deciphering its biological mechanism, atomic interactions, and functional sites. Human KRAS is a small GTPase consisting of 189 amino acids that cycles between active (GTP-bound) and inactive (GDP-bound) states. Computational tools such as AlphaFold DB and SWISS-MODEL generate accurate 3D atomic coordinates from primary amino acid sequences. Visualizing these structures using molecular graphics software like PyMOL permits clear identification of structural elements, including the central β -sheet core, amphipathic α -helices, flexible switch loops, and the catalytic GTP-binding pocket.

Objective

- Obtain the predicted 3D structure file (.pdb) for human KRAS (P01116).
- Load and render the structure in PyMOL using secondary structure cartoons and molecular surfaces.
- Identify key structural domains: central β -sheet, α -helices, P-loop, and switch regions.
- Interpret the relationship between KRAS 3D architecture and its GTPase function.

Results:





Interpretation

The 3D structural model of human KRAS clearly illustrates the spatial arrangement necessary for its role as a molecular switch. The central β -sheet provides a stable framework, while the P-loop forms a precise binding pocket for guanine nucleotides. The surface topology demonstrates how the active site cleft accommodates the phosphate tail of GTP, while Switch I and Switch II surround the pocket to adopt distinct conformations during signal transmission.

Conclusion

Three-dimensional visualization of human KRAS in PyMOL revealed a highly structured α/β domain optimized for nucleotide binding and catalysis. Identifying the P-loop and flexible switch regions highlights the structural basis of Ras GTPase signal transduction.